Algebra 1 Name____________________
Unit 9 Date_________________
Lesson 4 Homework Period__________

A new restaurant in Orlando, Florida determined that to maximize profits for the evening, they need to seat and serve 30 tables of guests. One evening, they were only able to seat and serve 16 tables, earning a profit of $\$352$ for the day. The restaurant is attempting to turn a maximum profit of $\$450$ for the day. The relationship between the restaurant's profit, y, and the number of tables served, x, can be modeled by a quadratic equation.

1. Identify the vertex and another point from the context.
   a. Vertex: (___, ___)
   b. Point: (___, ___)
2. Find the value of a for the quadratic equation modeling the restaurant's profit.
3. Write the quadratic equation that models the restaurant's profit in vertex form.
4. Rewrite the equation in standard form.

Lyon is attempting to build a fence for the largest possible rectangular play area at the local dog park. He can use up to 500 linear feet of fencing. He determines if he makes a rectangular play area with a length of 103 or 147 feet of fencing, the play area would cover $15{,}504$ square feet. To maximize the area, he can create a play area that is $15{,}625$ square feet with a length of 125 feet. The relationship between the length, x, and the area, y, of the rectangular play area can be modeled with a quadratic equation.

5. Identify the vertex and another point from the context.
   a. Vertex: (___, ________)
   b. Point: (___, _________)

6. Find the value of a for the quadratic equation that models the relationship.

7. Write the quadratic equation that models the relationship in vertex form.

The U.S. government tracks total emissions of greenhouse gases, in millions of metric tons from 2000 to 2010. After 4.4 years, greenhouse gas emissions reached its highest level for that period of time, at 7060 million metric tons. After 8.4 years, the greenhouse gas emissions had dropped to 6957.6 million metric tons. The relationship between years since 2000, x, and greenhouse gas emissions over this time period, y, can be modeled with a quadratic equation.

8. Identify the vertex and another point from the context.
   a. Vertex: (____, _________)
   b. Point: (____, __________)

9. Find the value of a for the quadratic equation modeling this relationship.

10. Write the quadratic equation that models this relationship in vertex form.